

II PUC MID-TERM EXAMINATION - 2025

CHEMISTRY (34)

Max Marks 70

Time 3.00 hrs.

Instructions :

1. The question paper has five parts. All parts are compulsory.
2. a) Part - A carries 20 marks. Each question carries 1 mark.
b) Part - B carries 3 marks. Each question carries 2 marks.
c) Part - C carries 15 marks. Each question carries 3 marks.
d) Part - D carries 20 marks. Each question carries 5 marks.
e) Part - E carries 3 marks. Each question carries 3 marks.
3. For Part - A questions, only the first written answers will be considered for evaluation.
4. Write balanced chemical equations and draw neat labelled diagrams and graphs wherever necessary.
5. Direct answers to the numerical problems without detailed steps and specific unit for final answer will not carry any marks.
6. Use log tables and simple calculator if necessary (Use of scientific calculator is not allowed).

PART - A

1. Select the correct option from the given choices :

$$15 \times 1 = 15$$

- 15x1=15
- The unit of concentration of solutions commonly used in medicine and pharmacy is
a) mass percentage
b) molarity
c) ppm
d) mass by volume percentage
 - The hydrogen halide which easily cleaves the carbon-oxygen (C-O) bond in ether is
a) HI
b) HBr
c) HCl
d) HF
 - Which reducing agent is used for the following conversion ?
 $\text{RCOOH} \longrightarrow \text{RCH}_2\text{OH}$
a) LiAlH_4
b) NaBH_4
c) $\text{K}_2\text{Cr}_2\text{O}_7$
d) KMnO_4
 - The substance used to determine the hardness of water using a simple titration is
a) Mg (EDTA)
b) Fe (EDTA)
c) Co (EDTA)
d) Na_2EDTA
 - The oxidation number of cobalt in $[\text{CoCl}_2(\text{en})_2]^+$ complex is
a) III
b) II
c) IV
d) VI
 - During the electrolysis of sodium chloride, the gas liberated at anode is
a) H_2
b) O_2
c) Cl_2
d) HCl
 - IUPAC name of ethylidene chloride is
a) 2,2 - dichloroethane
b) 1,2 - dichloroethane
c) 1,1 - dichloroethane
d) 2,1 - dichloroethane
 - The molecular formula of Freon-12 is
a) CF_4
b) CF_3Cl
c) CF_2Cl_2
d) CFCI_3
 - Statement I : NaCl and MgSO_4 have same value of constant "A" in plot of λ_m against $C^{1/2}$
Statement II : All electrolytes of a particular type have same value for "A". Choose the correct option.
a) Both the statements I and II are correct
b) Both the statements I and II are incorrect
c) Statement I is correct but statement II is incorrect
d) Statement I is incorrect but statement II is correct
 - K_H value for Ar(g) , $\text{CO}_2\text{(g)}$, HCHO(g) and $\text{CH}_4\text{(g)}$ are 40.3k bar, 1.67k bar, 1.83×10^{-5} k bar and 0.413 k bar respectively. The order of increasing solubility of these gases is
a) $\text{Ar} < \text{CH}_4 < \text{CO}_2 < \text{HCHO}$
b) $\text{Ar} < \text{CO}_2 < \text{CH}_4 < \text{HCHO}$
c) $\text{HCHO} < \text{CH}_4 < \text{CO}_2 < \text{Ar}$
d) $\text{HCHO} < \text{CO}_2 < \text{CH}_4 < \text{Ar}$
 - The order of boiling points of hydrogen halides is $\text{HCl} < \text{HBr} < \text{HI} < \text{HF}$. At STP, the hydrogen halide having maximum vapour pressure is
a) HCl
b) HBr
c) HI
d) HF

12. The SI unit of molar conductivity is

- a) S b) m^{-1} c) Sm^{-1} d) Sm^2mol^{-1}

13. When manganese dioxide is fused with KOH in air, it gives

- a) $KMnO_4$ b) Mn_2O_3 c) K_2MnO_4 d) MnO_2

14. Sufficient amount of 2-methylpropane-2-ol heated with 20% phosphoric acid at 358K gives main product 'X' with the elimination of water. t-Butyl alcohol undergoes dehydration when it is passed over heated copper at 573K and gives 'Y'. The boiling points of 'X' and 'Y' are

- a) Same b) 311.15K and 301.15K respectively
c) 301.15 K and 311.15K respectively d) 79°C and 93.5°C respectively

15. Match the following :

List I

- A. V_2O_5
B. $TiCl_4$ with $Al(CH_3)_3$
C. $PdCl_2$
D. Nickel complexes

List II

- i) Oxidation of ethyne to ethanol
ii) Polymerisation of alkynes
iii) Oxidation of SO_2 in the manufacture of sulphuric acid
iv) Manufacture of polyethylene

- a) (A) $\rightarrow$ (ii), (B) $\rightarrow$ (iv), (C) $\rightarrow$ (iii), (D) $\rightarrow$ (i) b) (A) $\rightarrow$ (iii), (B) $\rightarrow$ (iv), (C) $\rightarrow$ (i), (D) $\rightarrow$ (ii)
c) (A) $\rightarrow$ (i), (B) $\rightarrow$ (ii), (C) $\rightarrow$ (iv), (D) $\rightarrow$ (iii) d) (A) $\rightarrow$ (iv), (B) $\rightarrow$ (iii), (C) $\rightarrow$ (i), (D) $\rightarrow$ (ii)

II. Fill in the blanks by choosing the appropriate word from those given in the bracket:
(Phosgene, manganese, water, chromium, cis-platin, Zymase)

5x1=5

16. In the hydrogen-oxygen fuel cell, the only product is
17. Chloroform is slowly oxidised by air in the presence of light to form poisonous gas called
18. Glucose and fructose undergo fermentation in the presence of enzyme
19. 3d metal which shows highest oxidation state is
20. inhibits growth of tumours.

PART - B

III. Answer any three of the following. Each question carries two marks.

3x2=6

21. Write the equations for the steps in S_N1 mechanism of conversion of tertiary butyl bromide into tertiary alcohol.
22. Among alcohols and phenols which one is more acidic and why ?
23. State Kohlrausch law. Write one application of Kohlrausch law.
24. Calculate the magnetic moment of a divalent ion in aqueous solution, if its atomic number is 24.
25. Among O-nitrophenol and P-nitrophenol isomers, which is steam volatile ? Name the method used to separate those isomers.

PART - C

IV. Answer any three of the following. Each question carries three marks.

3x3=9

26. Explain the hybridisation, geometry and magnetic property of $[Co(NH_3)_6]^{3+}$ ion on the basis of valence bond theory (V.B.T.) (Atomic Number of Cobalt = 27)
27. State any three postulates of Werner's theory of coordination compounds.
28. Mention the lanthanoid product formed in the following reactions:
i) $Ln + C$ (at 2773K) $\rightarrow$
ii) $Ln + H_2O$ $\rightarrow$
iii) $Ln + S$ (Heat) $\rightarrow$
29. (a) Write the geometrical isomers of the compound. $[Pt(NH_3)_2Cl_2]$
(b) Write the IUPAC name of the following co-ordination compound : $[Pt(NH_3)_2Cl(NO_2)]$

30. When a chromite ore "A" is fused with sodium carbonate in free excess of air and the product is dissolved in water, a yellow solution of compound "B" is obtained. After treatment of this yellow solution with H_2SO_4 , compound "C" can be crystallised from the solution. When the compound "C" is treated with KCl, orange crystals of compound "D" is crystallised out. Write all reactions involved in the conversion of "A" to "D".

2x3=6

V. Answer any two of the following. Each question carries three marks.

31. Write 3 differences between Ideal and Non-Ideal solutions.
32. Regarding $H_2 - O_2$ fuel cell,
 i) Write electrode reaction at anode.
 ii) Name the catalyst incorporated into electrode.
 iii) Mention the electrolyte
33. Derive an expression relating molar mass of solution containing non-volatile solute and relative lowering of vapour pressure.
34. Plot a graph of molar conductivity Vs $C^{1/2}$ for strong electrolyte KCl and weak electrolyte CH_3COOH in solution. For strong electrolytes, write an equation that represents the increase in molar conductivity with dilution.

PART - D

VI. Answer any four of the following. Each question carries five marks.

4x5=20

35. a) Haloalkanes are less reactive towards nucleophilic substitution reactions. Give three reasons.
 b) Explain Finkelstein reaction with an example. (3+2)
36. a) Give the mechanism involved in the acid catalysed hydration of ethene to ethanol.
 b) What is Lucas reagent? How is Lucas reagent used to distinguish primary and tertiary alcohols? (3+2)
37. a) Explain Friedel-Crafts alkylation for chlorobenzene. Give equation.
 b) Explain Fittig reaction with an example.
 c) What are enantiomers? (2+2+1)
38. a) Complete the following equations:
 i) $C_6H_5Br + Mg \xrightarrow{\text{dry ether}} ?$
 ii) $CH_2 = CH_2 + Br_2 \xrightarrow{CCl_4} ?$
 iii)  $\xrightarrow[\text{ii) } H^+]{\text{i) } NaOH, 623K, 330 \text{ atm}} ?$
- b) What are freons? Give an example. (3+2)

39. A hydrocarbon 'A' having molecular formula C_9H_{12} is oxidized in the presence of air to 'B'. which on treating with dilute acid gives two commercial important organic compounds 'C' & 'G'. Organic compound 'C' gives a characteristic colour with aqueous $FeCl_3$. When 'C' is treated with dilute nitric acid at 298K yields isomers 'D' & 'E' where 'D' is less volatile than 'E', hence which are separated by steam distillation. Deduce the structure of A, B, C, D and E. (5)

40. a) How do you prepare methoxy ethane by Williamson's ether synthesis.
 b) What is the action of bromine in ethanoic acid on anisole? Give equation.
 c) Give the IUPAC name for $C_6H_5-O-C_2H_5$. (2+2+1)

PART - E (Problems)

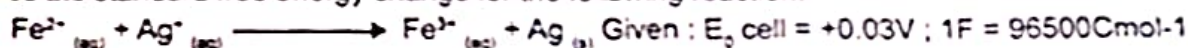
VII. Answer any three of the following. Each question carries three marks.

3x3=9

41. 200cm³ of an aqueous solution of a protein contains 1.26g of the protein. The osmotic pressure of such a solution at 300K is found to be 2057×10^{-3} bar. Calculate the molar mass of the protein. ($R = 0.0832 \text{ L bar mol}^{-1} \text{ K}^{-1}$)
42. 31g of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) is mixed with 186g of water. Calculate freezing point depression (K_f for water is $1.86 \text{ K Kg mol}^{-1}$)

43. The conductivity of 8 mol. L⁻¹ acetic acid is 40 S cm^{-1} . Calculate its dissociation constant, if limiting molar conductivity of acetic acid is $10 \text{ S cm}^2 \text{ mol}^{-1}$.

44. Calculate standard free energy change for the following reaction.



45. Calculate Henry's law constant when solubility of H_2S in water at STP is 0.195m. (Assume that H_2S exerts partial pressure of 0.987 bar)

46. The Galvanic cell is depicted as $\text{Zn (s) | Zn}^{2+} (\text{aq}) || \text{Cr}^{3+} (\text{aq}) | \text{Cr (s)}$; $E^\circ \text{ Zn}^{2+} | \text{Zn} = -0.76 \text{ V}$
 $E^\circ \text{ Cr}^{3+} | \text{Cr} = -0.74 \text{ V}$. Calculate the Equilibrium constant.

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